Newton’s rings experiment

Interference by division of amplitude

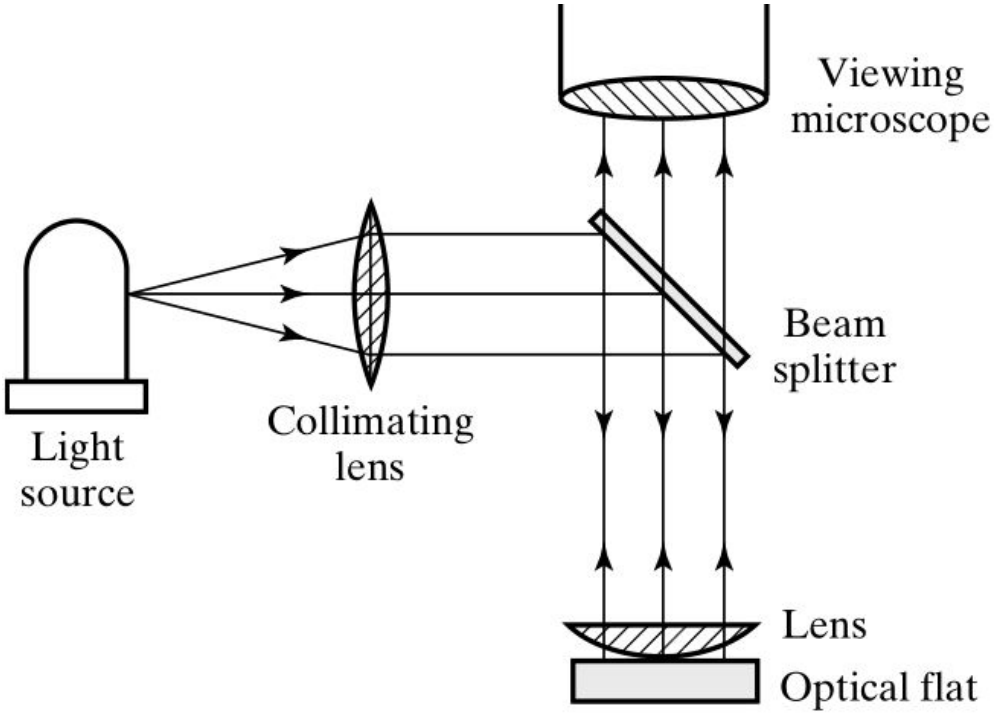
What is interference?

- Interference corresponds to superposition of two or more waves resulting in a spatially varying intensity distribution which is different from the sum of the irradiances of the individual components
- Take two waves with electric field amplitudes E_1 and E_2
- Corresponding irrandiances $I_1 = b|E_1|^2$ and $I_2 = b|E_2|^2$, b is a constant
- Resultant irradiance $I = b|E_1 + E_2|^2 \neq I_1 + I_2$

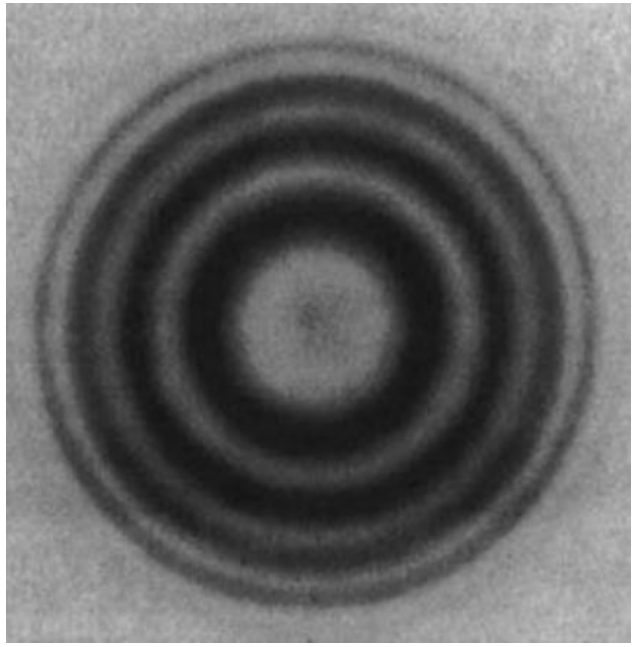
Examples:

- Interference of waves on water surface
- Standing waves on a string
- Interference of sound waves
- Interference of light waves (Observation of stable interference pattern in Young's double slit experiment confirmed the wave nature of light)

Newton’s rings experiment



Experimental scheme

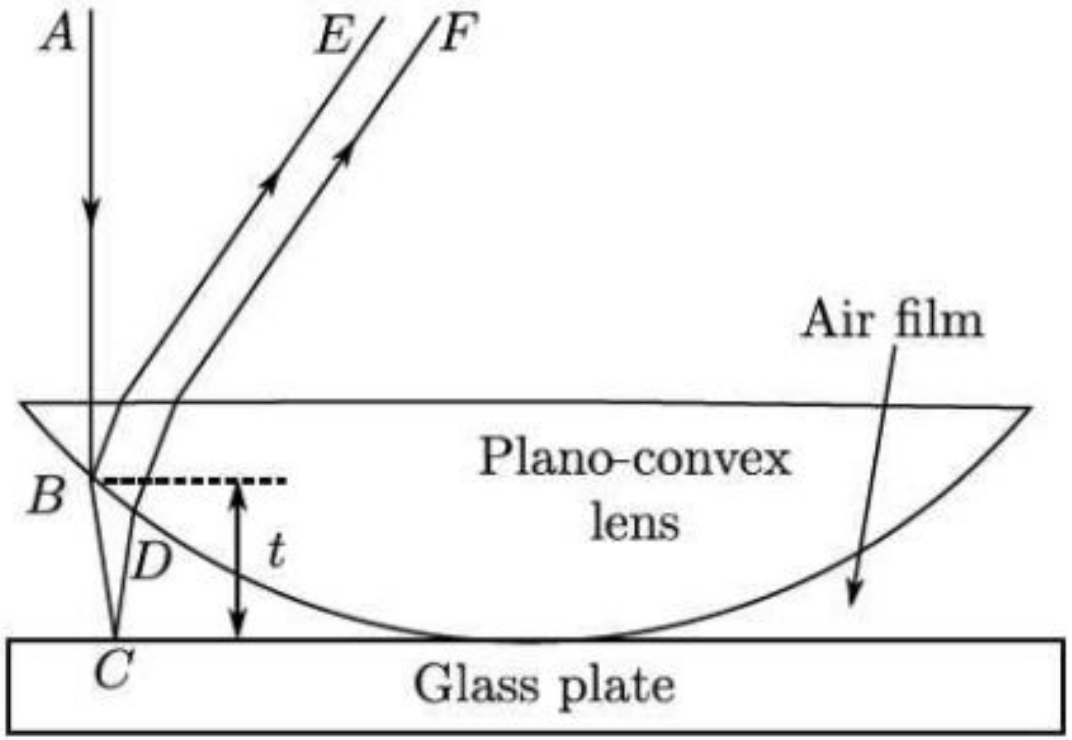


Ring pattern observed in reflection mode

1704 *“Newton’s rings” were first observed by Boyle and Hooke—they are named after Newton because he had given an explanation using the corpuscular model which was later found to be unsatisfactory.*

1802 *Thomas Young gave a satisfactory explanation of ‘Newton’s rings’ based on wave theory.*

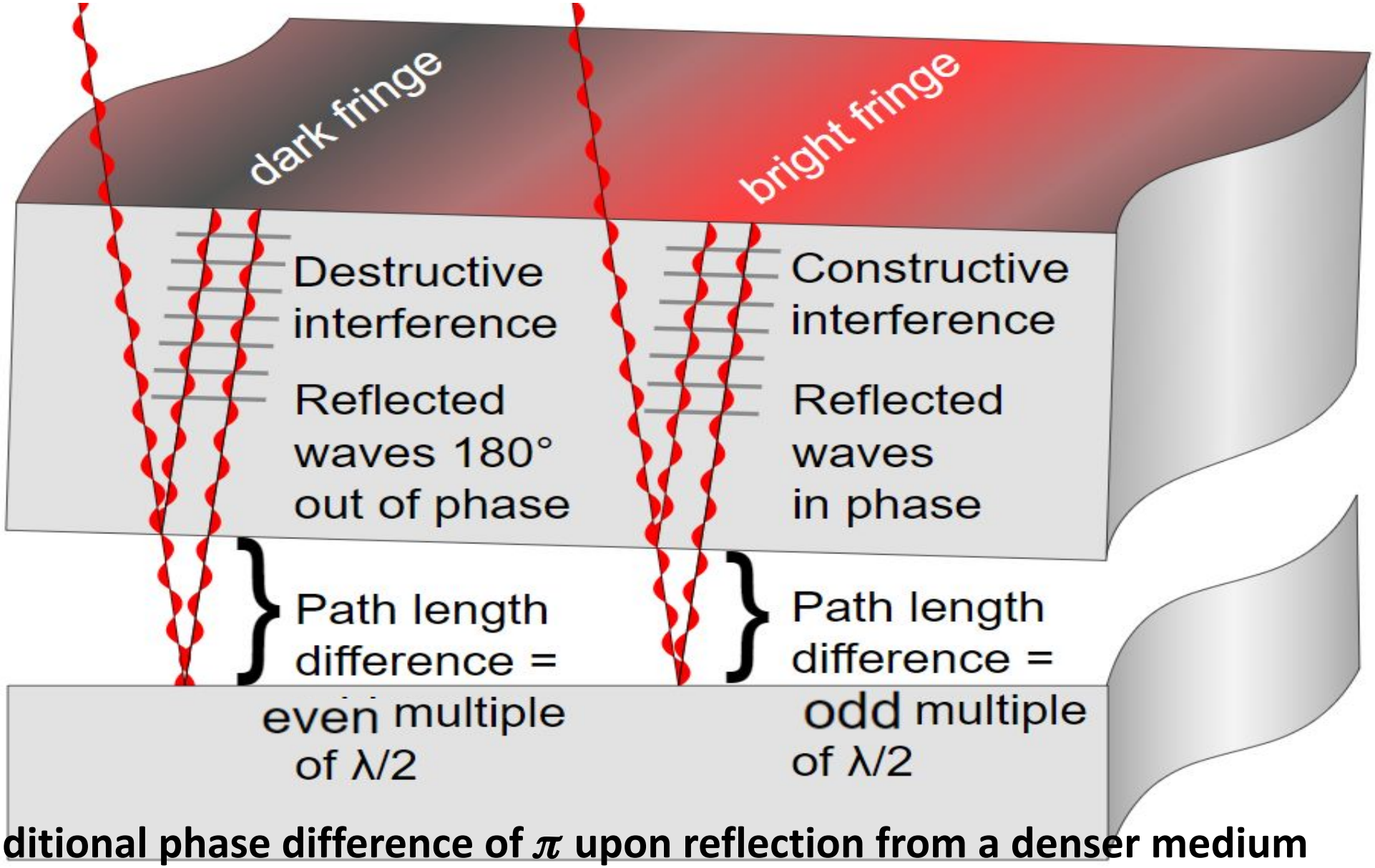
Close look at the schematic of Newton's rings interference expt.



- Constructive interference when
  $2\mu t = \left(n + \frac{1}{2}\right)\lambda;\ n = 0, 1, 2, \ldots.$
- Destructive interference when
  $2\mu t = n\lambda;\ n = 0, 1, 2, \ldots..$

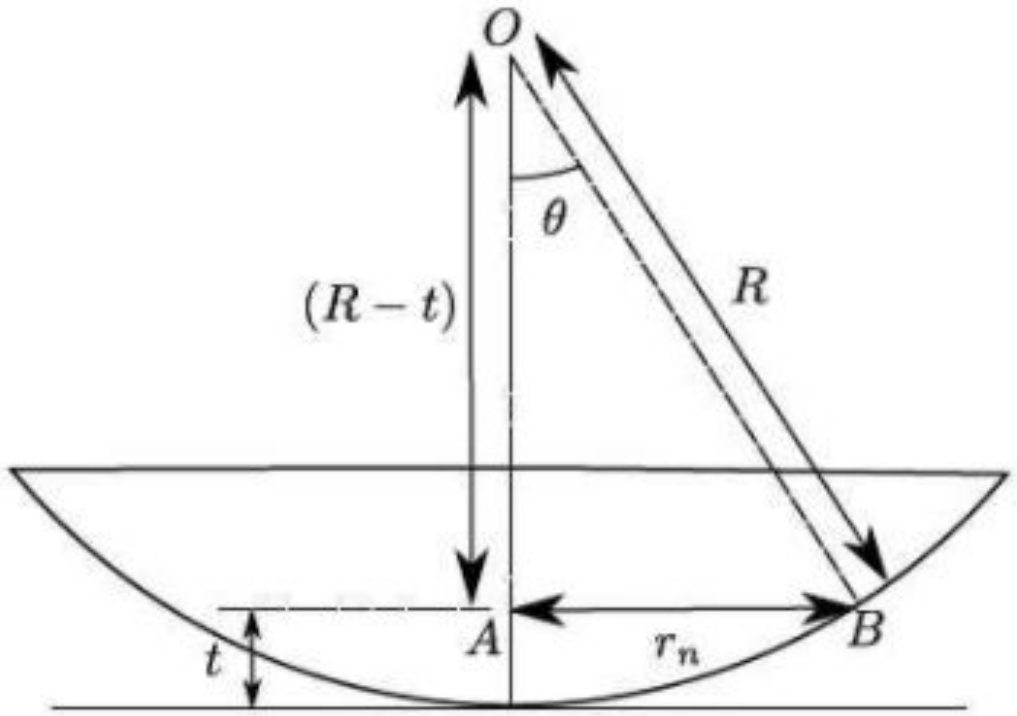
- Interference between rays reflected from the bottom surface of the lens and top surface of the glass plate
- Near normal incidence, observation is close to the point of contact between lens and glass plate, figure shows exaggerated view, the lens has large radius of curvature, thickness of the air film in between is very small (micron order)
- Ray-F which is reflected from the top surface of the glass plate has an additional phase of π due to reflection
- Path difference between these two rays depend on the radial distance from the point of contact between lens and glass plate

Schematic showing constructive and destructive interferences giving bright and dark fringes



Additional phase difference of π upon reflection from a denser medium

Calculate the thickness t of the air film at a radial distance r_n from the point of contact



- From ΔAOB, $R^2 = (R-t)^2 + r_n^2$; For $t \ll R$

$$2t \approx \frac{r_n^2}{R}$$

- Locus of points having equal path difference are concentric circles with point of contact as the center
- Radius of nth dark ring is $r_n = \sqrt{n\lambda R}$; $n = 0,1,2,3, \ldots \ldots$
- Radius increases as square root of natural numbers
- $r_{n+1} - r_n = \left(\sqrt{n+1} - \sqrt{n}\right)\sqrt{\lambda R}$
- Fringe width decreases with increasing n

$$r_{n+1}^2 - r_n^2 = \lambda R; \text{ For } n >\sim 10$$

$$r_{n+1} - r_n \approx \frac{\lambda R}{2r_n} = \frac{\sqrt{\lambda R}}{2\sqrt{n}}$$

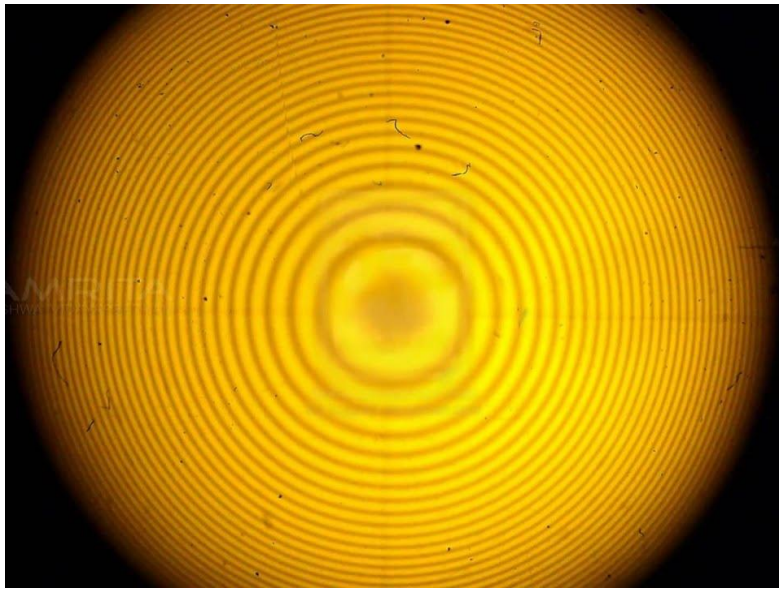
- If the space between the lens and glass plate is filled with a liquid of refractive index μ, the radius of the fringes decreases.
- $r_n = \sqrt{n\lambda R/\mu}$; $n = 0, 1, 2, \ldots$.

$$\text{For } n = 2,\ r_3 - r_2 \approx \frac{\sqrt{\lambda R}}{2\sqrt{2}} = 0.354\sqrt{\lambda R}$$

$$\text{For } n = 2,\ r_3 - r_2 = \left(\sqrt{3} - \sqrt{2}\right)\sqrt{\lambda R} = 0.318\sqrt{\lambda R}$$

$$\text{For } n = 10,\ r_{11} - r_{10} \approx \frac{\sqrt{\lambda R}}{2\sqrt{10}} = 0.158\sqrt{\lambda R}$$

$$\text{For } n = 10,\ r_{11} - r_{10} = \left(\sqrt{11} - \sqrt{10}\right)\sqrt{\lambda R} = 0.154\sqrt{\lambda R}$$



Intensity distribution: For two waves of equal intensities,

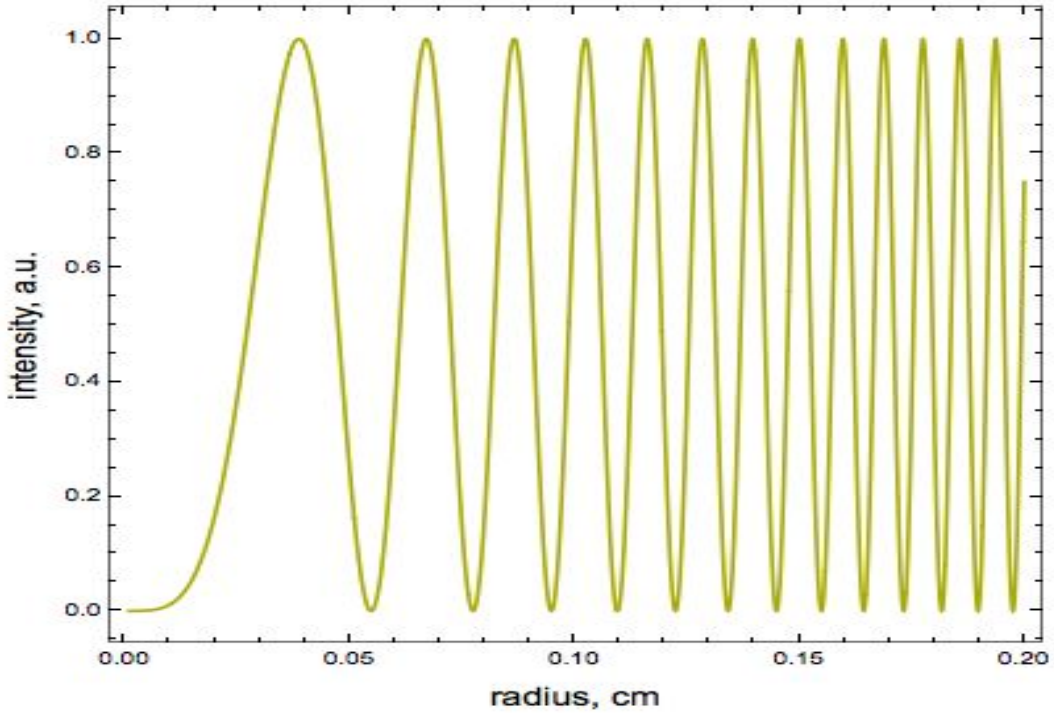
$$I = 4I_0 \cos^2\left(\frac{\delta}{2}\right),\ \delta = \text{ phase difference}$$

$$I = 4I_0 \cos^2\left(\frac{\pi\Delta}{\lambda}\right),\ \Delta = \text{ path difference } = \frac{2\pi}{\lambda}\delta$$

$$\text{For Newton's ring expt., } \Delta = 2t = \frac{r^2}{R}$$

Also, phase difference δ has an additional contribution of π for reflected rays

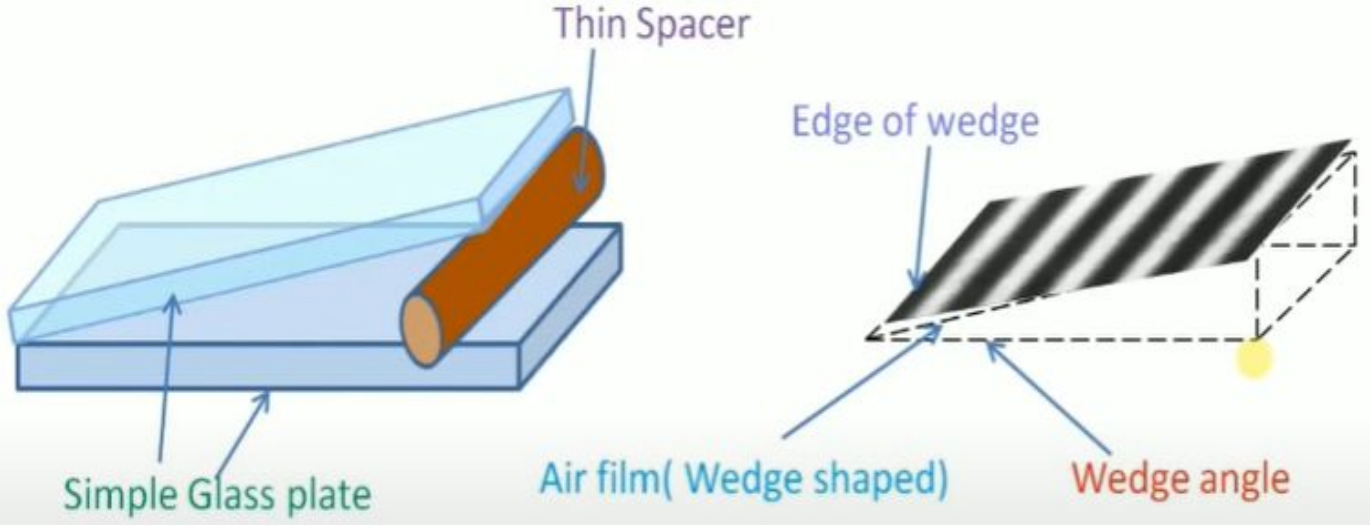
$$\text{So, } I = 4I_0 \cos^2\left(\frac{\pi r^2}{\lambda R} + \frac{\pi}{2}\right) = 4I_0 \sin^2\left(\frac{\pi r^2}{\lambda R}\right)$$

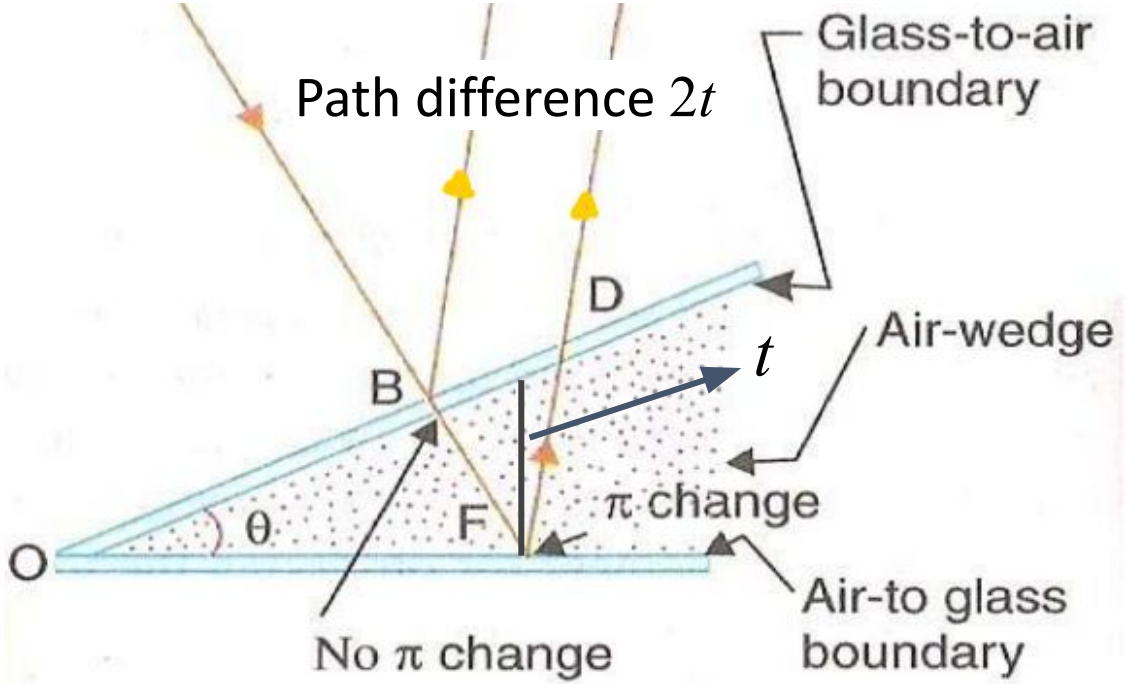


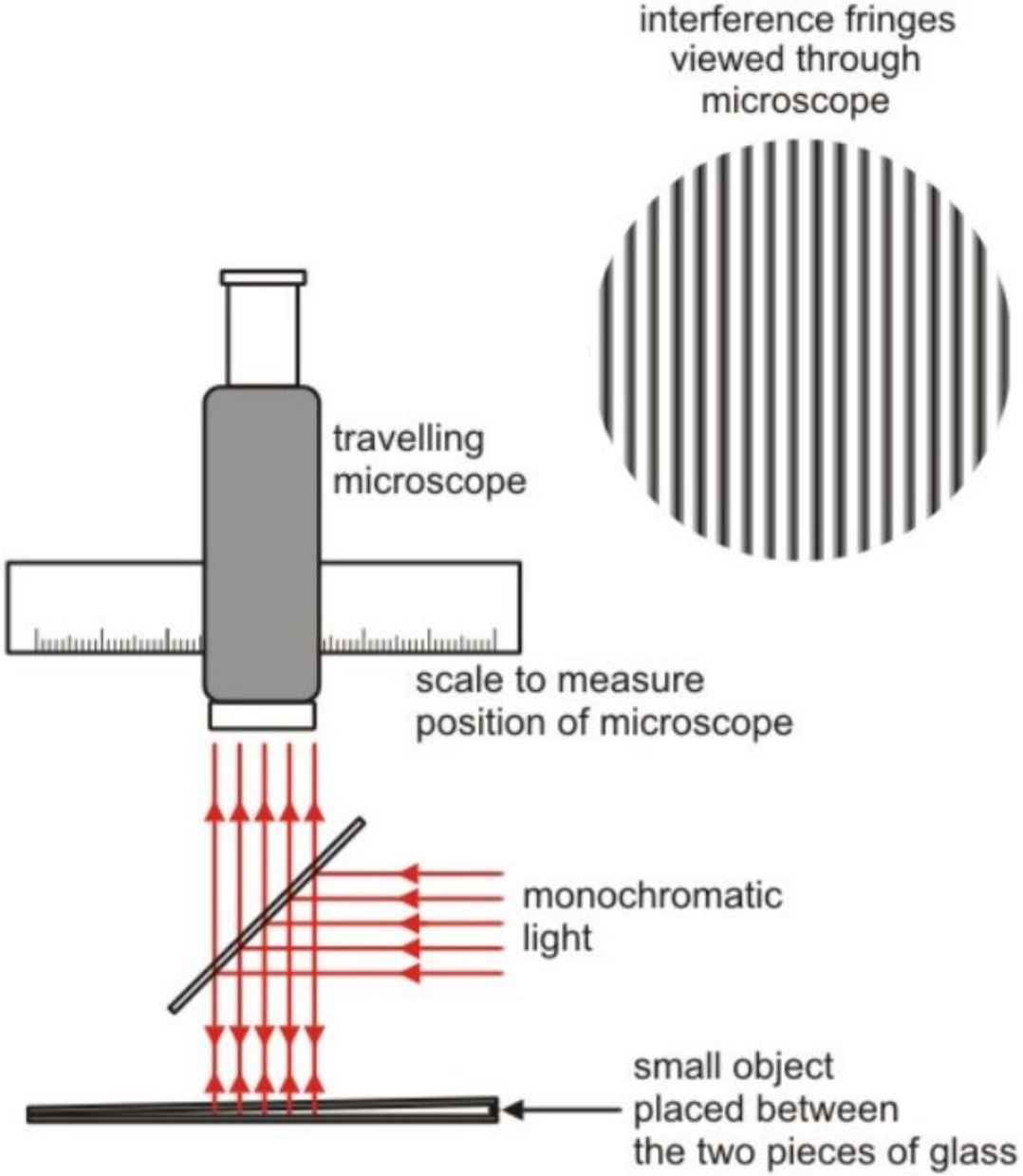
Central spot will be dark if the point of contact is perfect

- When we say that the point of contact is perfect, we still have a discontinuity of medium, the lens and glass plate are not in perfect contact at the atomic level, there is some air-gap whose thickness is sufficiently smaller than $\lambda/4$.
- In such case, due to the phase difference of π coming from reflection, there will be destructive interference and the central spot will be dark.
- In practice, there may be some undulation or dust particle, making the air gap at center larger than $\lambda/4$.
- In that case, the central spot may be bright.

Interference from two glass plate forming a wedge-shape

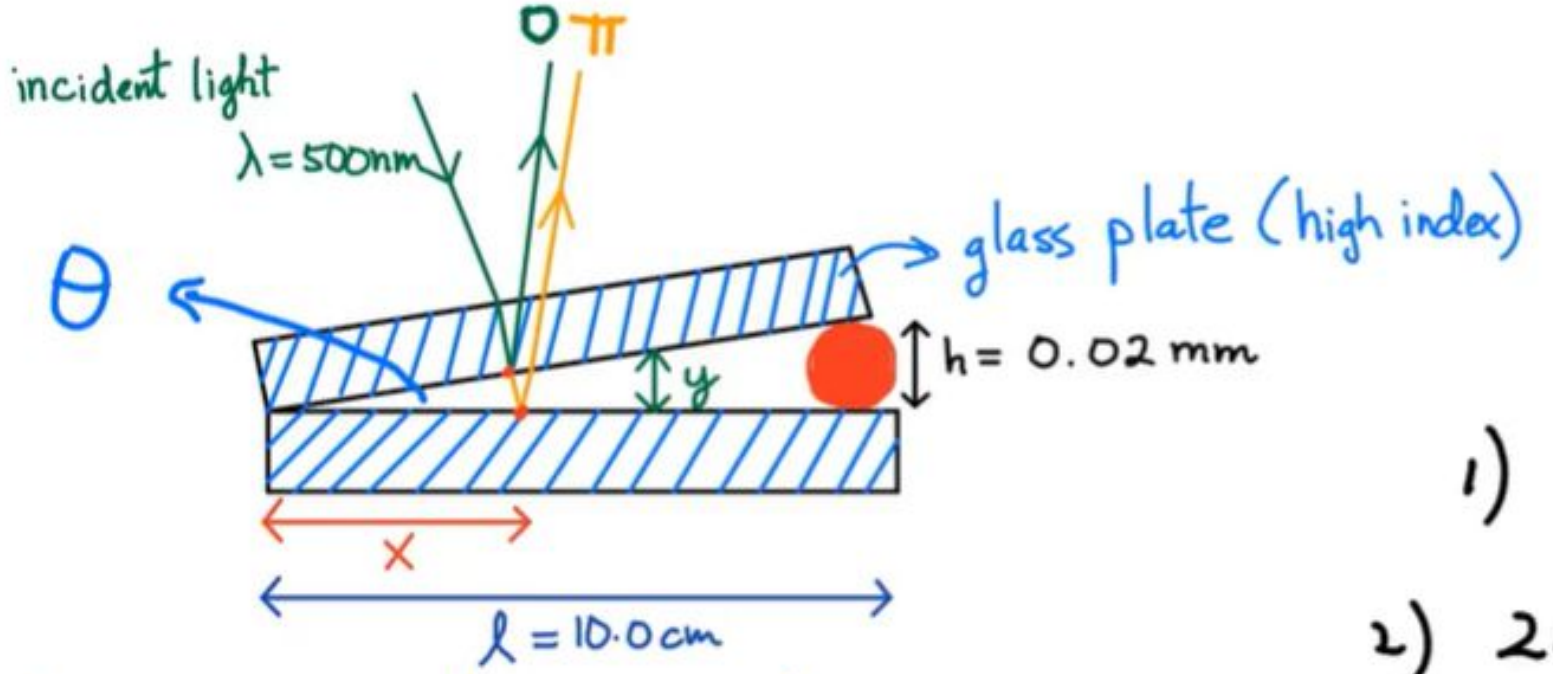


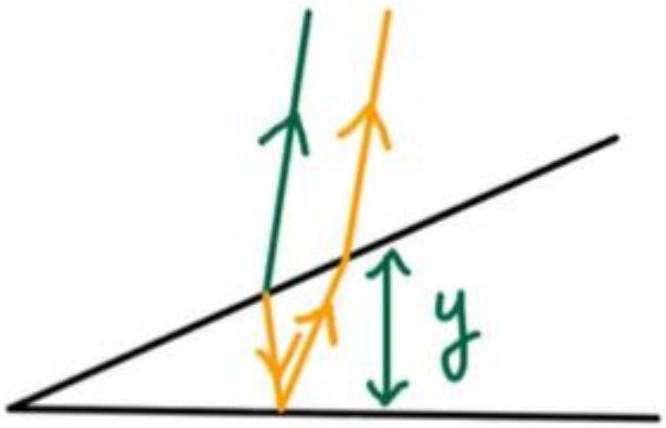




What is the spacing between interference fringes? $= 1.25\,mm$

$d = n\lambda \Rightarrow$ constructive





1) $2y = n\lambda$ (destructive)

2) $2y = \left(n + \frac{1}{2}\right)\lambda$ (constructive)

$n = 0, 1, 2, 3, \ldots,$

$\tan\theta = \frac{y}{x} = \frac{h}{\ell} \qquad y = \frac{h}{\ell}x$

1) $2\left(\frac{h}{\ell}x\right) = n\lambda \qquad x_n = n\left(\frac{\lambda \ell}{2h}\right) = (1.25\,mm)n$

↳ dark fringes

Application of Newton's rings apparatus

- Laboratory experiment for students training purpose
  - Determine the wavelength of light, given the radius of curvature of the lens
  - Determine the radius of curvature of the lens, given the wavelength of light

$$\lambda = \frac{D_{m+p}^2 - D_m^2}{4pR}$$

D_m is diameter of mth (bright or dark) ring

- A graph of D_{m+p}^2 vs p would be a straight line with slope $4\lambda R$ and intercept D_m^2

- Industrial application: test the surface quality (level of undulation) of optical elements

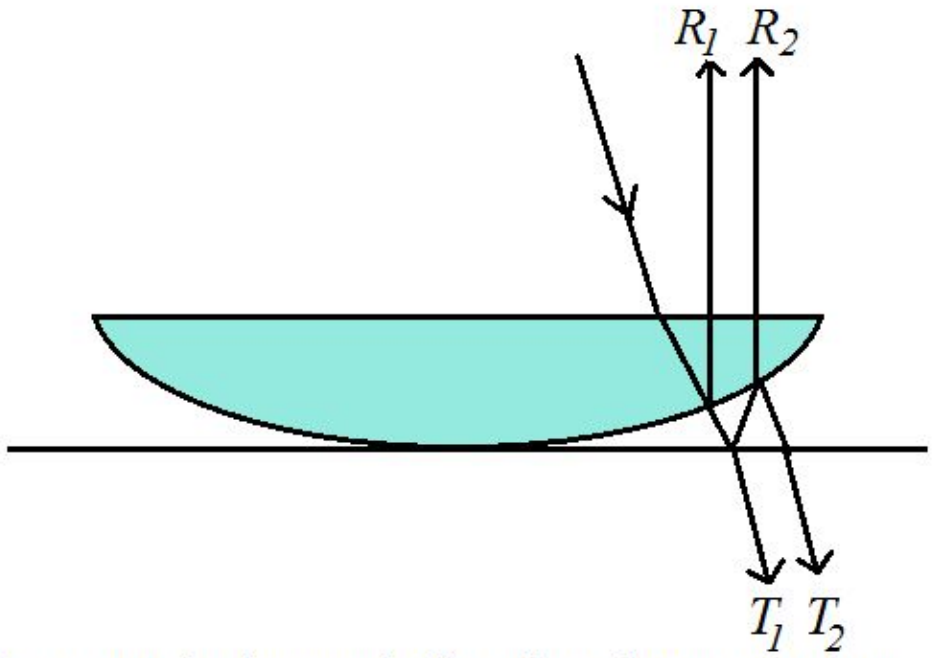
Testing flatness of a surface using Newton's rings like interference pattern

https://www.youtube.com/watch?v=8I5P-r4ogm4

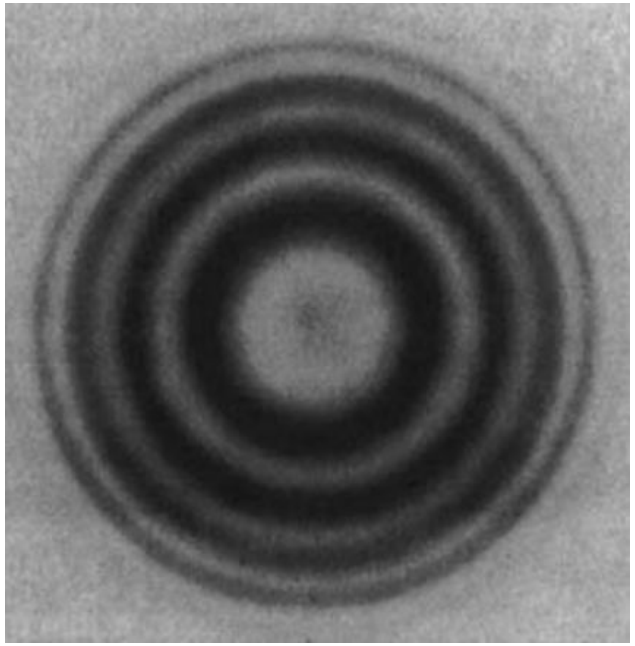
A few questions/concepts on Newton’s rings experiment

Newton's ring pattern observed both in reflection and transmission

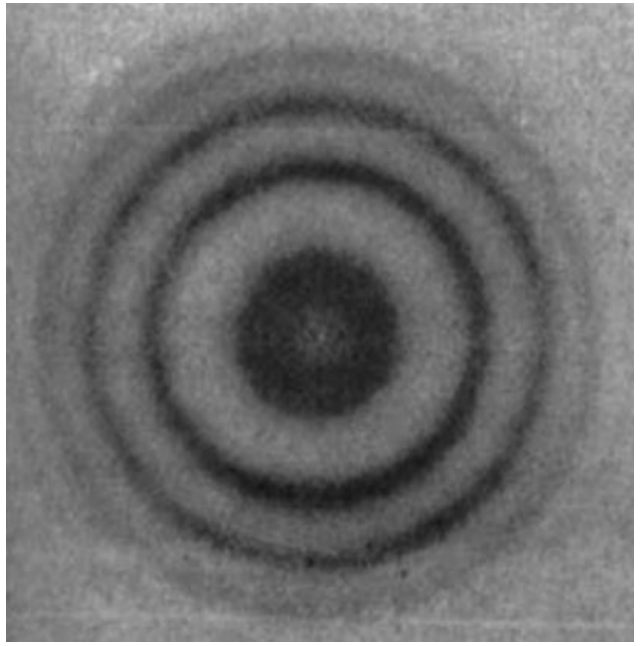
Similar to the reflection direction, ring pattern can also be observed in the transmission direction, reflection and transmission mode fringe patterns are complementary to each other; no net phase difference between two beams due to reflection; visibility can be poor as ray-T1 is stronger in intensity than ray-T2



Experimental scheme

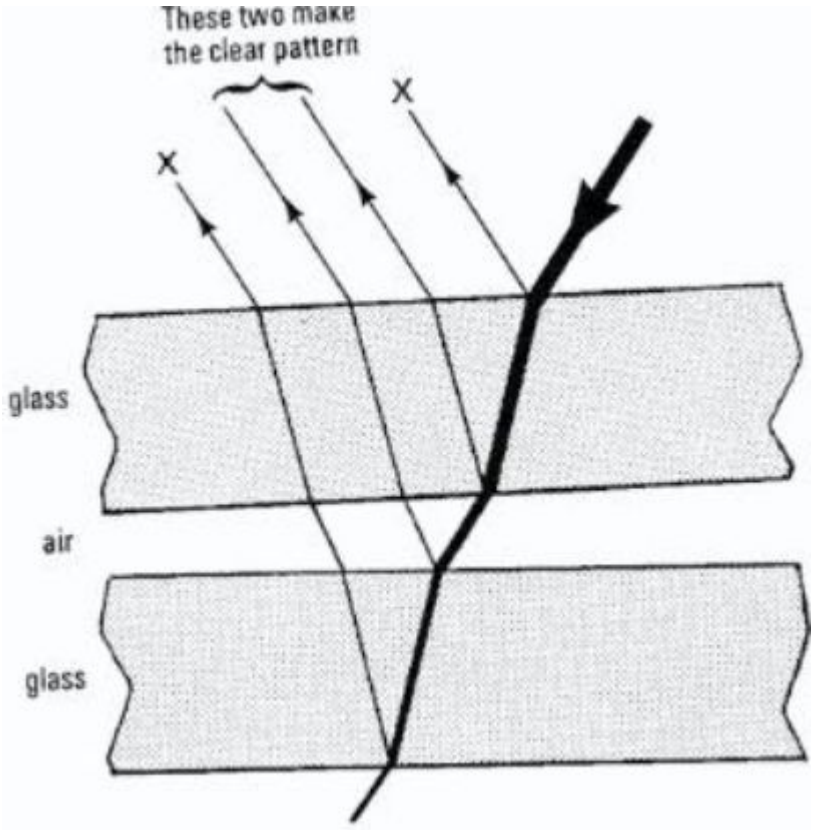


Reflection mode



Transmission mode

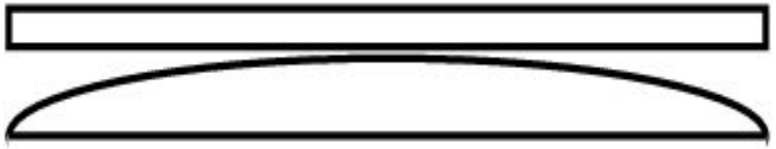
What happens to interference between rays reflected from surfaces other than that of the air film?



Only the air film between the lens and glass plate is very thin, and path difference between rays reflected from top and bottom of the air film is usually less than the coherence length of light used (often sodium vapour lamp is used as light source). Light reflected from other surfaces usually have path difference exceeding coherence length.

If laser light is used, interference fringes may be observed between other pairs of rays, depending on the coherence length of the laser light and the path difference between the pair of rays considered.

What if we put lens-plate combination upside down?

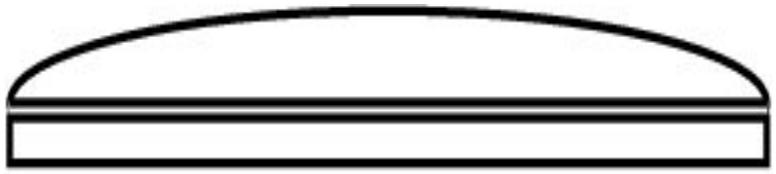


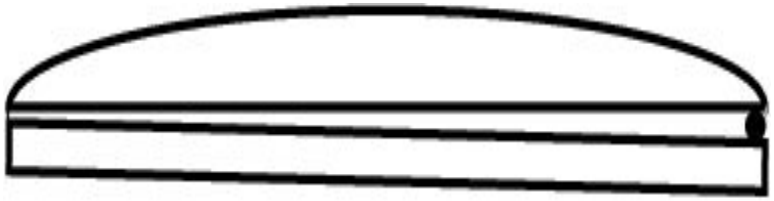
Practically no change in fringe pattern will be observed as we are looking at normal incidence, very close to the point of contact, and the air film is very thin as the lens has very large radius of curvature; we are neglecting any deviation of the beam from normal direction.

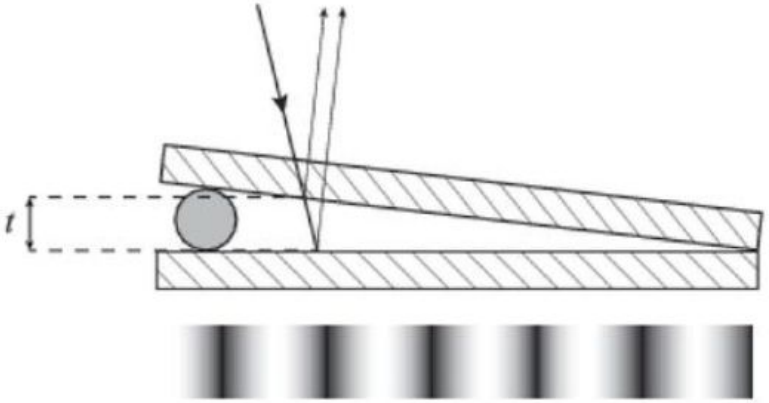
What if we put a biconvex lens on a flat glass plate?

Again we should not see any change in fringe pattern for the same reason as stated above

What if we put the flat face of the plano-convex lens on the glass plate?







In practice, the air gap can not be zero. There may be some undulation. Even if we neglect any undulation, there may be some dust particle trapped between the lens and glass plate leaving a wedge-shaped air gap. In that case we should see a series of parallel line fringes (as explained for the wedge-shaped air film between two glass plates).

What if we put a mirror in place of the glass plate?

Ray-2, which is reflected from the mirror, will have much higher intensity compared to ray-1, which is reflected from glass-air interface. As a result, the destructive interference (origin of dark fringe) will not be complete and fringe pattern will have poor contrast (visibility).

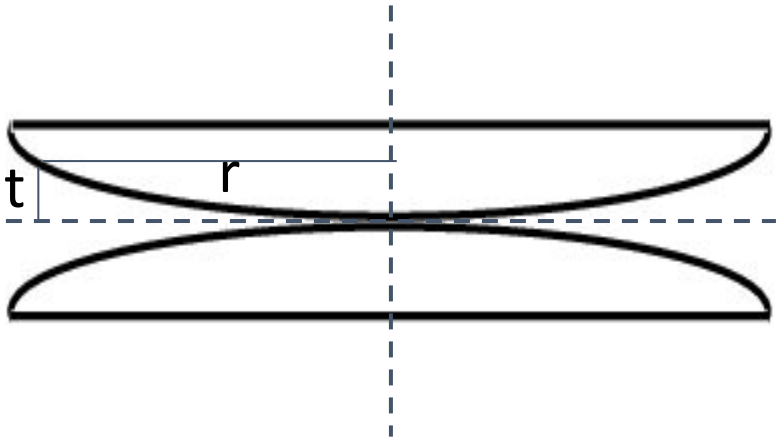
For interference of two rays with intensities $I_1 \text{ and } I_2$,

constructive interference (bright fringe) $\longrightarrow I_2 + I_1$ and

destructive interference (dark fringe) $\longrightarrow I_2 - I_1$

For $I_2 \gg I_1$, $I_2 + I_1 \approx I_2 - I_1 \approx I_2$ $\longrightarrow$ poor contrast (visibility)

What if we put two plano-convex lenses with curved faces back-to-back?

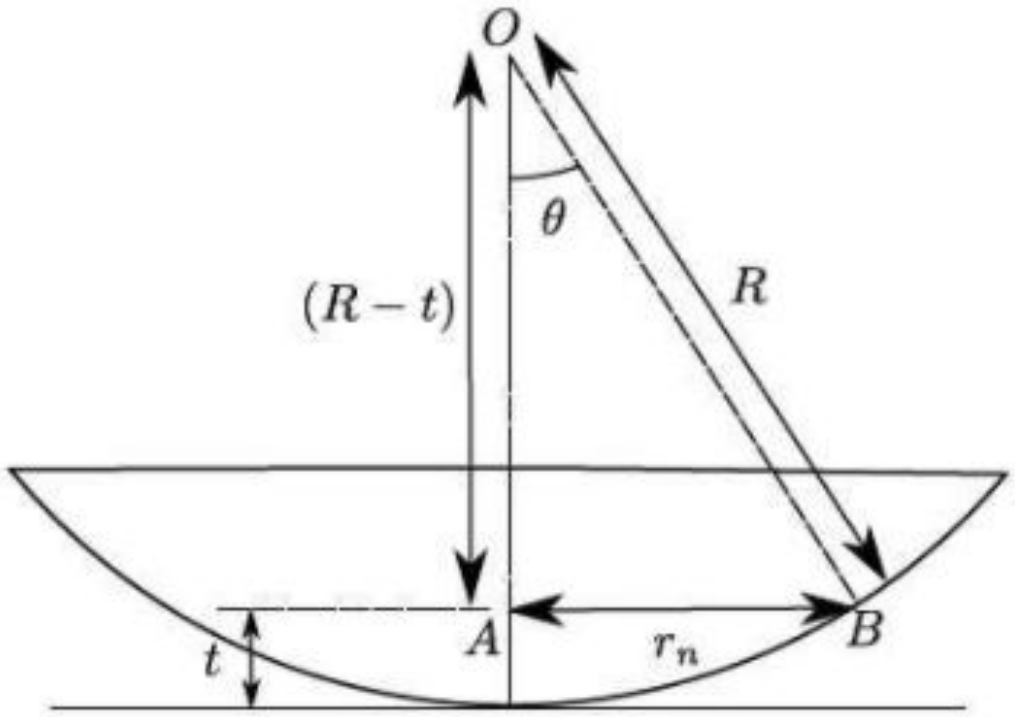


Path difference at a given radial position will increase; it will become double if the two lenses have same radius of curvature. Fringes will be more closely spaced and of reduced width.

What is we replace the air gap with a material of RI = μ

- If the space between the lens and glass plate is filled with a liquid of refractive index $\mu > 1$, the radius of the fringes decreases:

$$r_n = \sqrt{n\lambda R/\mu};\ n = 0, 1, 2, \ldots.$$ $\longrightarrow$ fringe pattern shrinks

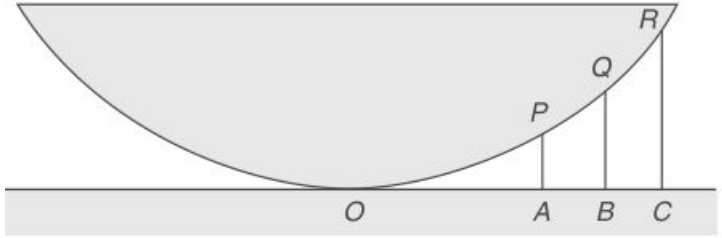


What if we slowly move the lens upward from the glass plate?

The fringe system would appear to shrink and it will be seen as if fringes are moving inwards and collapsing (disappearing) at the center.

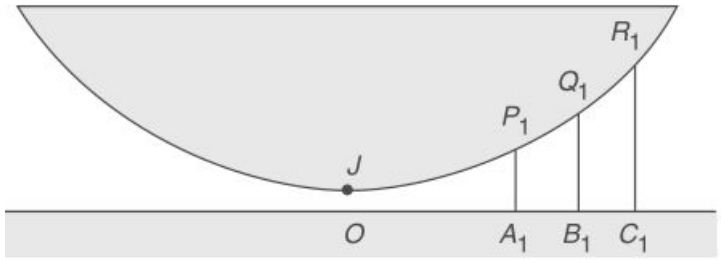
Let us illustrate it with an example.

Consider a standard Newton's ring setup. Assume light wavelength is 640 nm, the radius of curvature of the convex surface of the lens is 100 cm and the point of contact is perfect. We shall consider step by step what happens when the lens is moved upward slowly.



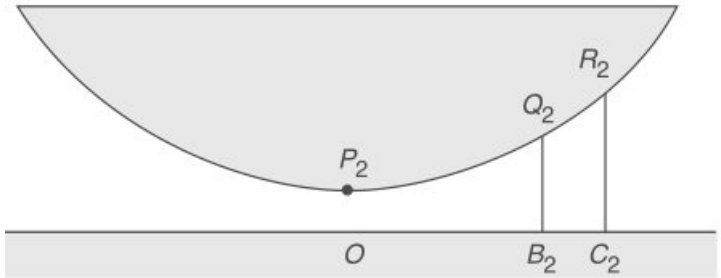
(a)

$PA = \lambda/2$
$QB = \lambda$
$RC = 3\lambda/2$



(b)

$JO = \lambda/4$
$P_1A_1 = \lambda/2$
$Q_1B_1 = \lambda$
$R_1C_1 = 3\lambda/2$



(c)

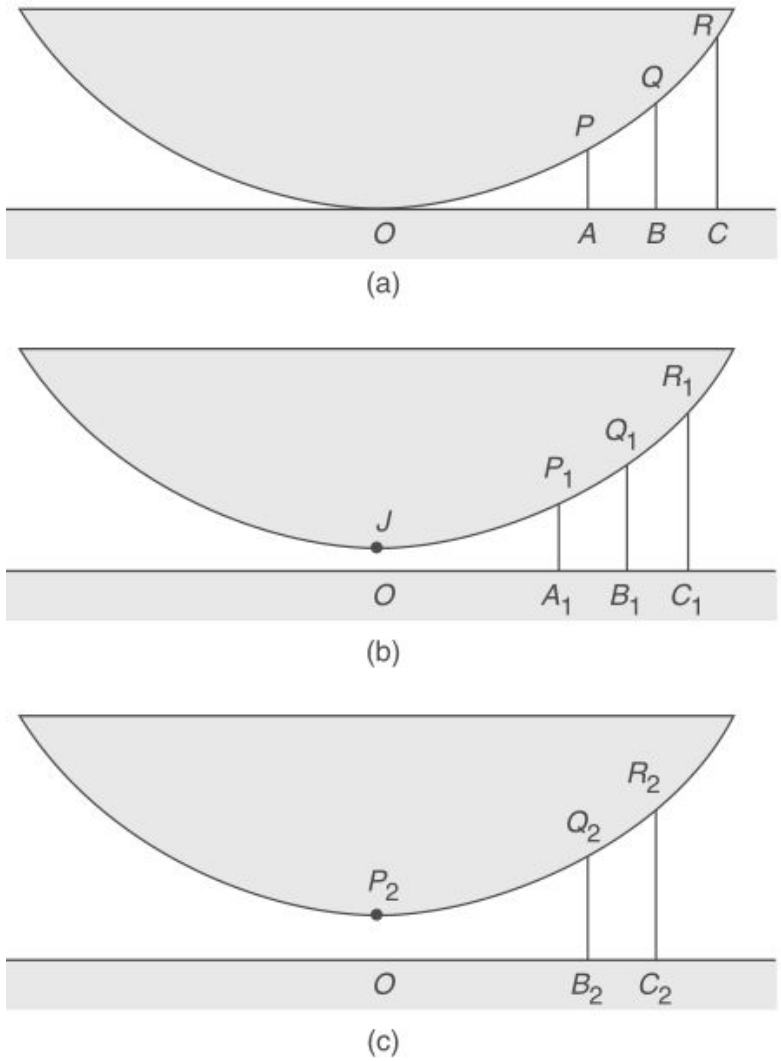
$P_2O = \lambda/2$
$Q_2B_2 = Q_1B_1 = QB = \lambda$
$R_2C_2 = R_1C_1 = RC = 3\lambda/2$

Since the point of contact is perfect, the central spot will be dark, the first dark ring will form at P where $PA = \lambda/2$, and the radius of this ring, OA, will be $\sqrt{\lambda R}$ (= 0.080 cm) –see Fig. (a). Similarly, the radius of the second dark ring will be $OB = \sqrt{2\lambda R}$ (= 0.113 cm). If we now raise the lens by $\frac{\lambda}{4}$(=1.6×10^{-5} cm) then $2t$ corresponding to the central spot would be $\lambda/2$ and instead of the dark spot at the centre we will now have a bright spot. The radii of the first and the second dark rings will be

$$OA_1 = \left(\frac{1}{2}\lambda R\right)^{1/2} = 0.0566 \text{ cm}$$

and

$$OB_1 = \left(\frac{3}{2}\lambda R\right)^{1/2} = 0.098 \text{ cm}$$



$PA = \lambda/2$
$QB = \lambda$
$RC = 3\lambda/2$

respectively [see Fig. (b)]. If the lens is further moved by $\lambda/4$ (see Fig. (c)], then the first dark ring collapses to the center and the central spot will be dark. The ring which was originally at Q now shifts to Q_2; similarly the ring at R [Fig. (a)] collapses to R_2 [Fig. (c)].

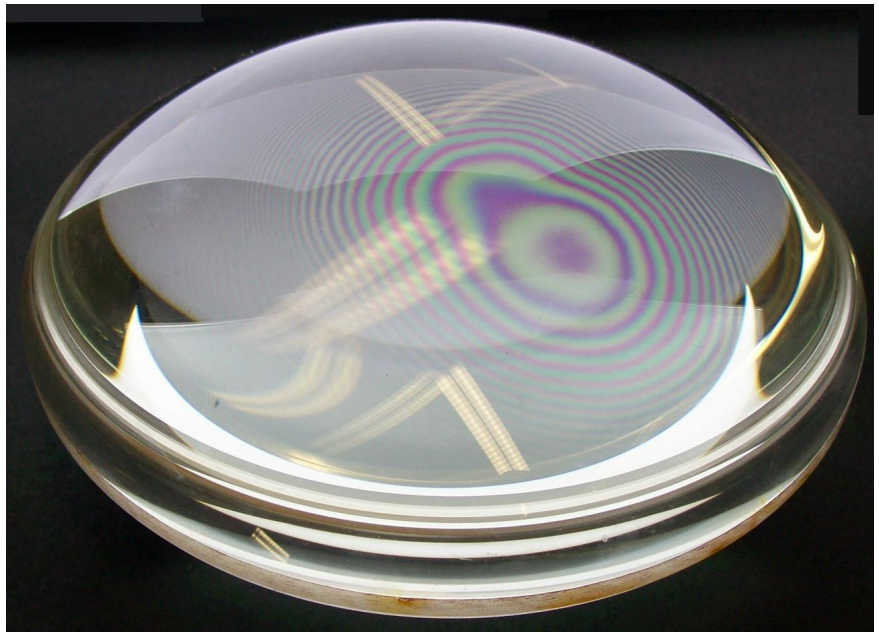
$JO = \lambda/4$
$P_1A_1 = \lambda/2$
$Q_1B_1 = \lambda$
$R_1C_1 = 3\lambda/2$

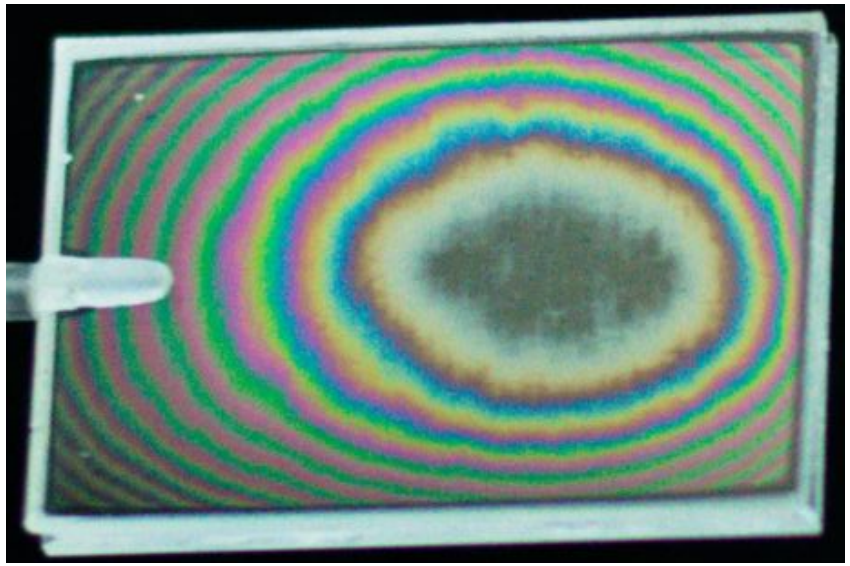
Thus, as the lens is moved upward the rings collapse to the centre. Hence if we can measure the distance by which the lens is moved upward and also count the number of dark spots that have collapsed to the center, we can determine the wavelength. For example, in the present case, if the lens is moved by 6.4×10^{-3} cm, 200 rings will collapse to the center.

$P_2O = \lambda/2$
$Q_2B_2 = Q_1B_1 = QB = \lambda$
$R_2C_2 = R_1C_1 = RC = 3\lambda/2$

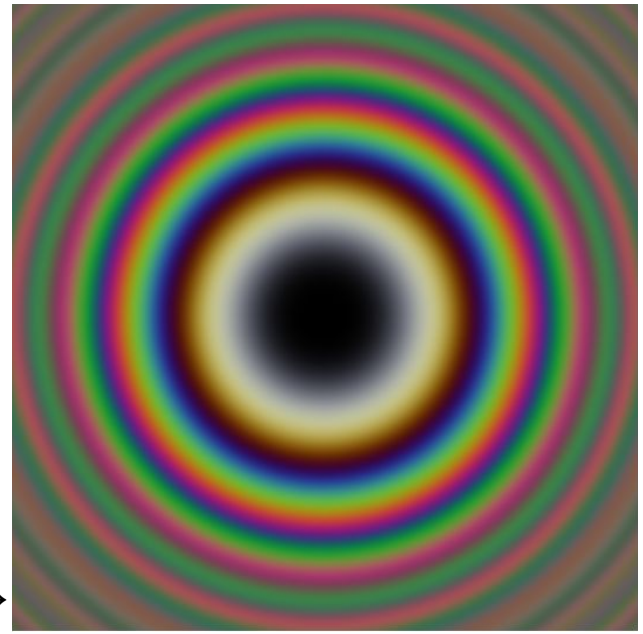
What if we use a white light source instead of a monochromatic one

- One would see colored fringes, as the condition of maximum for a given order will be satisfied at slightly different radial position for different wavelengths. If the the point of contact is perfect, then the central spot should be dark. Away from the center, the fringes will be colored.
- Fringe pattern will be washed out at a distance from the center, as the maxima from one color will tend to overlap with the minima from another color.





Simulation with 12 colors to simulate white light→



What if we use a color filter to view white light fringes?

For fringes created by white light source, if we use a color filter to view the fringe pattern, we can see the fringes at one color which is allowed to pass through the particular color filter. This will look similar to the fringe pattern created by a monochromatic source of that color.

Why do we usually not use a laser source for Newton's ring expt.?

It is possible to observe Newton's ring using laser light sources. However, as the laser has long coherence length, rays reflected from surfaces other than that of the air film may also produce interference fringes. So, there may be overlap of different fringe systems. Measurements become difficult.

Newton's rings: A classroom demonstration with a He–Ne laser

A. F. Leung and Jiyeon Ester Lee
Department of Physics and Astronomy, California State University, Long Beach, California 90840

(Received 27 December 1989; accepted for publication 19 July 1990)

Am. J. Phys. 59, 662–664 (1991)
https://doi.org/10.1119/1.16792

What if two nearby wavelengths are present in the light source?

- For location near the center, the bright and dark fringes for the two wavelengths will, respectively, almost coincide
- But away from the center, the fringe pattern from two wavelength will be shifted from one another and at some point a bright fringe of one color will coincide with a dark fringe from another color (half fringe shift). The fringe pattern will be blurred around that radial distance.
- At a larger radius, the fringe pattern will be seen again. But again at another radius bright fringe of one color will be with a dark fringe from another color (one and half fringe shift). The fringe pattern will again be blurred around that radial distance. This process will be repeated.

What if two nearby wavelengths are present in the light source?

- Let us see this through an example.
- Consider a standard Newton's ring setup illuminated by sodium vapor light which predominantly consists of 2 wavelengths: λ_1 = 589 nm and λ_2 = 589.6 nm
- Since the two wavelengths are very close to each other, near the center (for small order of fringe), bright and dark fringes corresponding to λ_1 will almost coincide with respective bright and dark fringes corresponding to λ_2.
- Let us see this with a calculation: let us calculate the radii of 10th bright and dark fringes for these two wavelengths.
- Radius of mth dark fringe = $\sqrt{m\lambda R}$ and that of mth bright fringe = $\sqrt{(m+0.5)\lambda R}$

- Take $R = 100$ cm. For λ_1 = 589 nm, radius of 10th dark ring = 0.242693 cm and that of 10th bright ring = 0.248687 cm
- For λ_2 = 589.6 nm, radius of 10th dark ring = 0.242817 cm and that of 10th bright ring = 0.248813 cm
- Fringe shift between λ_1 and λ_2 for m = 10 is 0.000125 cm
- Fringe width for m^{th} order fring is $\sim 0.5\sqrt{\lambda_{av}R/m}$. For m = 10, it is 0.012138 cm
- Fringe shift is about 1% of fringe width
- So, for small m, bright and dark fringes arising from these two wavelengths nearly superpose on one another, respectively.
- Let us now see what happens for large m, say, m = 300.

- For λ_1 = 589 nm, radius of 300th dark ring = 1.32929 cm and that of 300th bright ring = 1.33039 cm
- For λ_2 = 589.6 nm, radius of 300th dark ring = 1.32996 cm and that of 300th bright ring = 1.33107 cm
- Fringe shift between λ_1 and λ_2 for m = 300 is 0.000675 cm
- Fringe width for m = 300 is 0.002216 cm
- Fringe shift is about 30.5% of fringe width
- So, the bright and dark fringes from one wavelength are considerably shifted relative to the corresponding fringes from other wavelength for large m.
- Proceeding this way, there will be a situation when m^{th} dark fringe from λ_1 will coincide with m^{th} bright fringe from λ_2. Then the fringe pattern will washed out (uniform illumination instead of bright and dark fringes) around that value of m. Say, this corresponds to the air-gap thickness t and radial position r.

- So, under this condition we have: $2t = m\lambda_1 = (m + 0.5)\lambda_2$

$$\Rightarrow \frac{2t}{\lambda_2} - \frac{2t}{\lambda_1} = \frac{1}{2} \;\Rightarrow\; 2t\frac{\lambda_1 - \lambda_2}{\lambda_1\lambda_2} = \frac{1}{2} \;\Rightarrow\; t \approx \frac{1}{4}\frac{\lambda_{av}^2}{\Delta\lambda} \;\text{ where } \Delta\lambda = \lambda_1 - \lambda_2$$

and for $\lambda_1 \cong \lambda_2$, $\lambda_1\lambda_2 \cong \lambda^2_{av}$.

- For λ_1 = 589 nm and λ_2 = 589.6 nm, we get $t \cong$ 0.01447 cm. This corresponds to $m \cong$ 491, and radial position r ≅ 1.70118 cm
- Moving further beyond this radial position, the relative shift will increase beyond half-fringe width and fringes will be visible again.
- Vanishing of fringes will be repeated whenever the condition $\frac{2t}{\lambda_2} - \frac{2t}{\lambda_1} = p + \frac{1}{2}, p = 0, 1, 2, \ldots\ldots$ is fulfilled.
- Instead of moving in radial direction, same situation can be simulated by moving the lens upward and thereby increasing t i.e. the path difference
- This can give a method of estimating coherence length